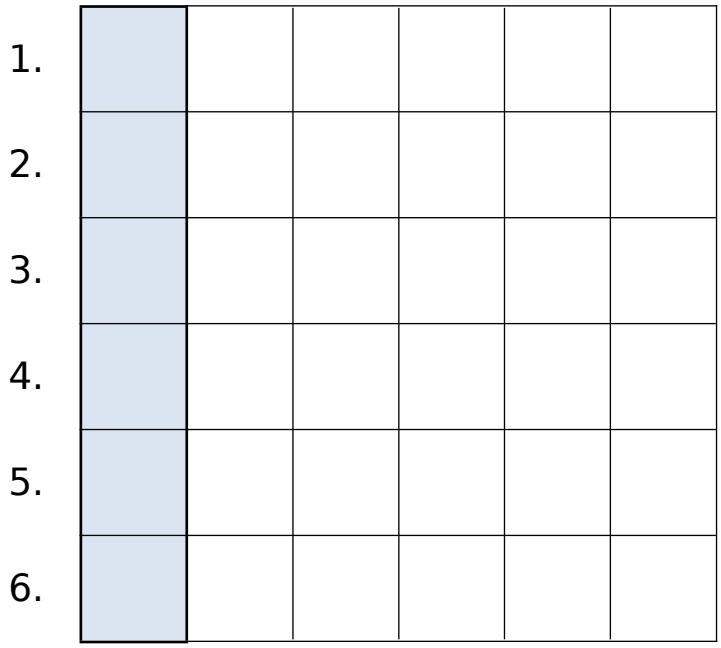
Puzzle 1

Can you solve this chemistry puzzle?

Answer the six clues across to fill each horizontal line in the square grid below. On completion ‘1 down’ should be a ‘horizontal row’ in itself!



Across

1. The nucleus of a hydrogen atom only contains this particle.

2. For a positive ion to form an electron must ..... from the atom.

3. Potassium metal..... violently with water.

4. A solid member of the halogen family!

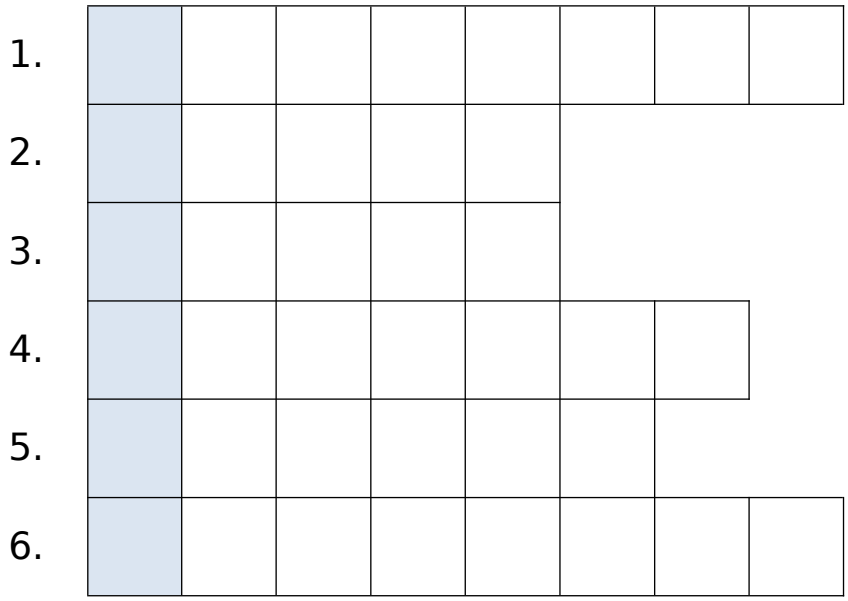
5. The compounds formed when oxygen reacts chemically with elements.

6. Group II elements form ions with a ..... positive charge.

Puzzle 2

Answer the six clues to find six **non-metallic elements**.

On completion ‘1 down’ should be another non-metallic element and is found in all living things.



Across

1. This non-metal is a pale yellow-green poisonous gas. (8)

2. An inert gas used in light bulbs. (5)

3. A radioactive inert gas. (5)

4. The halogen which is a liquid at room temperature. (7)

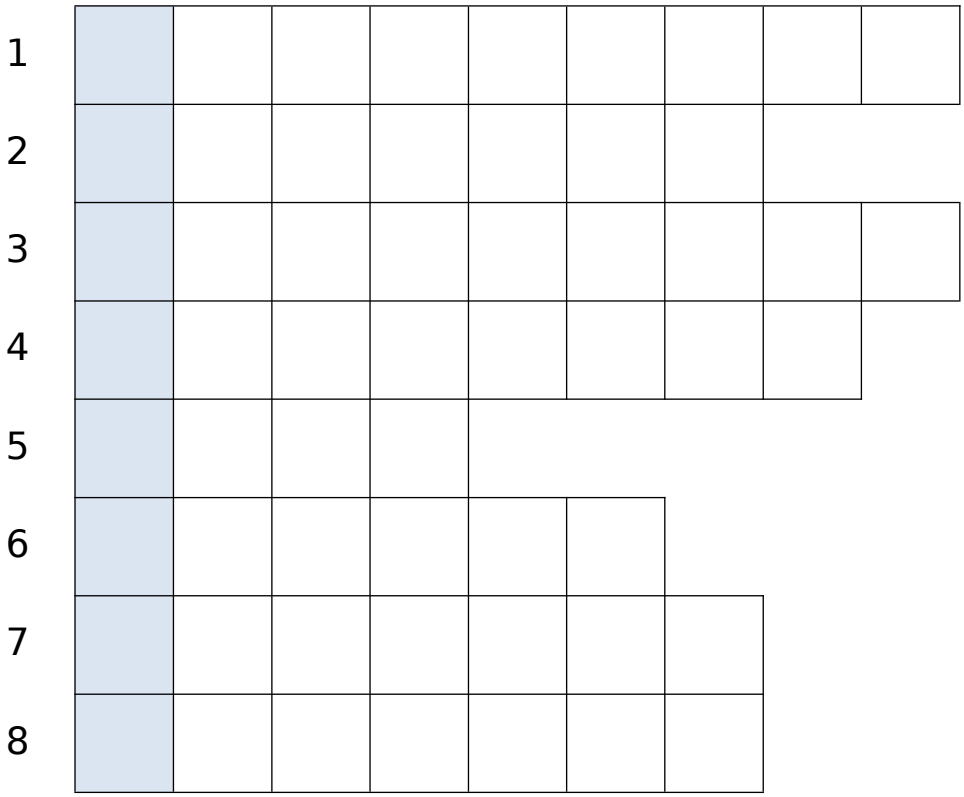
5. This non-metal is produced during photosynthesis. (6)

6. This non-metal has a triple covalent bond between the atoms in its molecule. (8)

Puzzle 3

Answer each of the eight clues about metals.

When you have finished ‘1 down’ will form a very precious metal.



1. A radioactive metal named after the dwarf planet in our solar system. (9)

2. The first alkali metal in the group. (7)

3. A low density metal used for making window frames and cooking foil. (9)

4. A metal with a very high melting point, used as a filament in electric light bulbs. (8)

5. A metal with the chemical symbol Fe. (4)

6. This metal is mixed with copper to form an alloy which is used for coinage. (6)

7. A radioactive metal whose 235-isotope is used in nuclear power stations. (7)

8. The only liquid metal at room temperature. (7)

© www.teachitscience.co.uk 2018

Answers

1.

P	R	O	T	O	N
E	S	C	A	P	E
R	E	A	C	T	S
I	O	D	I	N	E
O	X	I	D	E	S
D	O	U	B	L	E

2.

C	H	L	O	R	I	N	E
A	R	G	O	N			
R	A	D	O	N			
B	R	O	M	I	N	E	
O	X	Y	G	E	N		
N	I	T	R	O	G	E	N

3.

P	L	U	T	O	N	I	U	M
L	I	T	H	I	U	M		
A	L	U	M	I	N	I	U	M
T	U	N	G	S	T	E	N	
I	R	O	N					
N	I	C	K	E	L			
U	R	A	N	I	U	M		
M	E	R	C	U	R	Y		